# Nikolaos Tsilivis

ntsilivis@ttic.edu, +1-347-721-1666
6045 S Kenwood Ave, Chicago, IL 60637

**Research Interests**

Foundations of Machine Learning, Robust Machine Learning, Applications of Machine Learning in Natural Sciences.

**Employment**

**Toyota Technological Institute at Chicago** — Chicago, IL, USA
Research Assistant Professor — Sep 2026 -

**Fundamental AI Research (FAIR), Meta** — NYC, NY, USA
Visiting Researcher — Sep 2024 - Sep 2025

**Education**

**New York University** — NYC, NY, USA
Ph.D. in Data Science — Sep 2021 - Aug 2026
Ph.D. Advisor: Julia Kempe

**École Normale Supérieure de Paris** — Paris, France
Visiting Ph.D. student — Apr - Jun 2024, Oct - Dec 2025
Hosts: Julia Kempe and Bruno Loureiro

**Toyota Technological Institute at Chicago** — Chicago, IL, USA
Visiting Ph.D. student — Jan - Mar 2024
Host: Nathan Srebro

**Kempner Institute, Harvard University** — Cambridge, MA, USA
Visiting Ph.D. student — Sep - Dec 2023
Hosts: Boaz Barak and Cengiz Pehlevan

**National Technical University of Athens** — Athens, Greece
Diploma in Electrical and Computer Engineering (B.Sc. & M.Sc.) — Sep 2014 - Jul 2021
Thesis Advisor: Petros Maragos

**KTH Royal Institute of Technology** — Stockholm, Sweden
Exchange Studies — Jan - Jun 2018

**Publications**

*Conferences*

22. Learning through Internalization
**N. Tsilivis**, N. Joshi, M. Medvedev, J. Kempe, N. Srebro
NeurIPS 2026

21. How reinforcement learning after next-token prediction facilitates learning
**N. Tsilivis**, E. Malach, K. Ullrich, J. Kempe
ICLR 2026
EurIPS 2025 Workshop on Principles of Generative Modeling ⋆*Contributed talk*⋆

20. OpenApps: Simulating Environment Variations to Measure UI Agent Reliability
K. Ullrich, J. Su, C. Shi, A. Subramonian, A. Bar, I. Evtimov, **N. Tsilivis**, R. Balestriero, J. Kempe, M. Ibrahim
ICLR 2026 ⋆*Oral*⋆

19. Flavors of Margin: Implicit Bias of Steepest Descent in Homogeneous Neural Networks
**N. Tsilivis**, G. Vardi, J. Kempe
ICLR 2025

18. The Evolution of Statistical Induction Heads: In-Context Learning Markov Chains
E. Edelman$^\star$, **N. Tsilivis**$^\star$, B. L. Edelman, E. Malach, S. Goel
NeurIPS 2024

17. The Price of Implicit Bias in Adversarially Robust Generalization
**N. Tsilivis**, N. Frank, N. Srebro, J. Kempe
NeurIPS 2024

16. What Can the Neural Tangent Kernel Tell Us About Adversarial Robustness?
**N. Tsilivis**, J. Kempe
NeurIPS 2022

15. Sparsity in Max-Plus Algebra and Applications in Multivariate Convex Regression
**N. Tsilivis**, A. Tsiamis, P. Maragos
ICASSP 2021

14. Sparse Approximate Solutions to Max-Plus Equations
**N. Tsilivis**, A. Tsiamis, P. Maragos
Discrete Geometry and Mathematical Morphology 2021 $\star$*Invited to the special issue*$\star$

*Journals*

13. Flavors of Margin: Implicit Bias of Steepest Descent in Homogeneous Neural Networks (extended version)
**N. Tsilivis**, E. Gronich, J. Kempe, G. Vardi
Journal of Machine Learning Research 27

12. On the Geometry of Regularization in Adversarial Training: High-Dimensional Asymptotics and Generalization Bounds
M. Vilucchio, **N. Tsilivis**, B. Loureiro, J. Kempe
TMLR 2026

11. Kernels, data & physics
F. Cagnetta, D. Oliveira, M. Sabanayagam, **N. Tsilivis**, J. Kempe ($\alpha$–$\beta$ order)
Journal of Statistical Mechanics: Theory and Experiment 2024

10. On the Robustness of Neural Collapse and the Neural Collapse of Robustness
J. Su, Y. S. Zhang, **N. Tsilivis**, J. Kempe
TMLR 2024

9. Attacking Bayes: On the Adversarial Robustness of Bayesian Neural Networks
Y. Feng, T. Rudner, **N. Tsilivis**, J. Kempe
TMLR 2023 $\star$*Reproducibility certification*$\star$

8. Toward a Sparsity Theory on Weighted Lattices
**N. Tsilivis**, A. Tsiamis, P. Maragos
Journal of Mathematical Imaging and Vision 2022

*Workshops*

7. Learning to Think from Multiple Thinkers
N. Joshi, R. Magen, N. Srebro, **N. Tsilivis**, G. Vardi ($\alpha$–$\beta$ order)
NeurIPS 2026 Workshop on Transitioning from Pre-Training to Post-Training

6. A Tale of Two Circuits: Grokking as Competition of Sparse and Dense Subnetworks
W. Merrill$^\star$, **N. Tsilivis**$^\star$, A. Shukla
ICLR 2023 Workshop on Mathematical and Empirical Understanding of Foundation Models

5. Can we achieve robustness from data alone?
**N. Tsilivis**, J. Su, J. Kempe
ICML 2022 Workshop on New Frontiers in Adversarial Machine Learning

4. Adversarial Noise Injection for Learned Turbulence Simulations
J. Su, J. Kempe, D. Fielding, **N. Tsilivis**, M. Cranmer, S. Ho
NeurIPS 2022 Workshop on Machine Learning and the Physical Sciences

*Preprints*

2. From Target Shift to Covariate Shift: How RL Post-Training Enables Transfer to New Tasks
E. Cornacchia$^\star$, **N. Tsilivis**$^\star$, J. Kempe, B. Loureiro, arXiv 2026

1. Extracting Finite Automata from RNNs Using State Merging
W. Merrill$^\star$, **N. Tsilivis**$^\star$, arXiv 2022

**Awards**

*G-Research PhD Prize, Courant Institute School of Mathematics, Computing, and Data Science* (2025-2026): One of two recipients recognized for the best PhD dissertations in machine learning across mathematics, computer science, and data science.

*Coefficient Giving Grant - 88,000$* (2026-2027): Academic grant for project: "Theoretical Foundations of Robust Autoregressive Learning as a Means of Designing Safe and Robust Models".

*Meta AI Research Grant* (2024-2025): Covers full tuition and stipend as a part of the Meta AI Mentorship Program for one academic year.

*Gerondelis Foundation Grant* (2024): Awarded to Greek students pursuing graduate studies in the United States.

*Center for Data Science Fellowship* (2021-2026): Covers tuition and living expenses for 5 years.

*Thomaideio Award (Publications)* (2021): Awarded to undergraduate students of the National Technical University of Athens who published a research paper before their graduation.

**Teaching**

Teaching Assistant for DS-GA 1005, NYU: *Inference and Representation* (2025)
Instructor: Joan Bruna

Teaching Assistant for DS-GA 2003, NYU: *Introduction to Data Science for Ph.D. Students* (2022)
Instructors: Kyunghyun Cho, Cristina Savin, Julia Kempe

Co-authored lecture notes for Julia Kempe's lectures at the 2022 Les Houches Summer School on Statistical Physics and Machine Learning

**Invited talks**

Aspen Center for Physics, Aspen Meeting on Foundation Models
Title: *Learning through Internalization*

ICTP, Youth in High-Dimensions: Recent Progress in Machine Learning, High-Dimensional Statistics and Inference (2026)
Title: *Learning through Internalization* (video link)

EPFL, Lausanne Event on Machine Learning and Neural Network Theory (2026)
Title: *How reinforcement learning after next-token prediction facilitates learning*

Toyota Technological Institute at Chicago, Young Researcher Seminar Series (2026)
Title: *How reinforcement learning after next-token prediction facilitates learning*

Center for Computational Mathematics, Flatiron Institute (2026)
Title: *How reinforcement learning after next-token prediction facilitates learning*

EurIPS 2025, Workshop on Principles of Generative Modeling (2025)
Title: *How reinforcement learning after next-token prediction facilitates learning*

Stanford University, Simons Collaboration on the Physics of Learning and Neural Computation Kickoff Workshop (2025)
Title: *How reinforcement learning after next-token prediction facilitates learning*

Microsoft Research NYC (2025)
Title: *How reinforcement learning after next-token prediction facilitates learning*

École Normale Supérieure de Paris, CSD Seminar (2025)
Title: *How reinforcement learning after next-token prediction facilitates learning*

Toyota Technological Institute at Chicago, NSF TRIPODS Workshop (2024)
Title: *The Price of Implicit Bias in Robust ML*

Flatiron Institute, Center for Computational Mathematics (2024)
Title: *The Price of Implicit Bias in Robust ML*

University of California, Los Angeles, IPAM, Workshop II: Theory and Practice of Deep Learning (2024)
Title: *The Price of Implicit Bias in Robust ML* (video link)

Harvard University, ML Foundations group (2023)
Title: *The best algorithm for adversarial training*

New York University, J. Bruna's group (2023)
Title: *Discontinuous Deep Learning, Grokking & More*

University of California, Irvine - GoalLab (2022)
Title: *Lazy Optimization Regimes in Deep Learning*

New York University, CDS PhD seminar (2022)
Title: *What Can The Neural Tangent Kernel Tell Us About Adversarial Robustness?*

**Service**

Mentoring & Community Service:

- CDS & NYU GSAS Ph.D. Mentoring Programs: Paired with incoming students to provide guidance and support (2024-2025)
- Supervision of undergraduate & master students:
  - Aman Shukla (CDS, NYU), 2022-2023: optimization dynamics in neural networks

  - Ya Shi Zhang (Courant, NYU), 2022-2023: adversarial robustness of neural networks
  - Sergey Sedov (CDS, NYU), 2024-2025: efficient methods for reasoning in language models
  - Iason Kalogiannis (ECE, NTUA), 2025-2026: mean-field analysis of neural networks with ranking layers

- Member of the Student Inclusion and Belonging Advisory Board at CDS, NYU (Spring 2025)
- NeurIPS 2024 Volunteer.

Reviewing: ALT 2027, NeurIPS 2024 (**Top reviewer**), 2026, ICLR 2023, 2025 (**Notable reviewer**), AISTATS 2025, Physical Review Letters, TMLR, UniReps Workshop (NeurIPS 2023), Re-Align Workshop (ICLR 2024), HiLD Workshop (ICML 2024), SciFor4DL Workshop (NeurIPS 2024), M3L Workshop (NeurIPS 2024), MOSS Workshop (ICML 2025), PriGM Workshop (EurIPS 2025).

Area Chair: ICLR 2027, ICML 2027.

**Programming Skills**

Languages: Python, C/C++, Java, SML, Prolog

Other: PyTorch, JAX

**Languages**

Greek (native), English (fluent)